The 0/0 Atlas

A Complete Catalog of Indeterminate Form in Mathematics

The L.O.R.E. Collaboration

Puronbo Laboratory

2026-08-18

arXiv: 2026.PunoCalculus

THE 0/0 ATLAS

A Complete Classification of Indeterminate Forms Across Mathematics

Authors: The L.O.R.E. Collaboration

Date: 2026-08-18

Repository: Puronbo/Law-Of-Repulsive-Emanation

Classification: Reference atlas

Abstract

We present a complete classification of 55 instances of the indeterminate form 0/0 across nine branches of mathematics. Each instance is classified by (i) the branch, (ii) the mechanism (Probe, Index, Vanishing Rate, Critical Phenomenon, or Conservation), (iii) the exact 0/0 form, (iv) the removable value, (v) the theorem it encodes, and (vi) the computational verification. We provide cross-reference tables, a decision tree for classifying new instances, and a catalog of open problems.

Part I: Definitions

1.1 The indeterminate form

Definition. An expression f/g is an indeterminate form of type 0/0 at x? if:

1. f and g are both defined in a punctured neighborhood of x?
2. lim_{x->x?} f(x) = 0 and lim_{x->x?} g(x) = 0

3. The limit lim_{x->x?} f(x)/g(x) exists and is finite

The removable value is the limit.

Remark. Condition 3 is not automatic -- the limit might not exist (e.g., f(x)/g(x) = sin(1/x) as x -> 0). When it does exist, the singularity is removable.

1.2 The five mechanisms

Mechanism I (Probe). f/g = 1 where defined, but f and g both vanish at isolated points. The removable value at each zero tests whether f and g are "the same."

Mechanism II (Index). The 0/0 arises from an integral of the form ? (something)/(something else) around a zero. The removable value is an integer -- a winding number, index, or multiplicity.

Mechanism III (Vanishing Rate). A quantity h(t) vanishes at t = 0. The ratio h(t)/t^n is 0/0. The removable value is the leading Taylor coefficient.

Mechanism IV (Critical Phenomenon). At a phase transition, two divergent or vanishing quantities form a 0/0 or 0xinf. The removable value is a critical amplitude.

Mechanism V (Conservation). A conserved quantity is the ratio of two quantities that both vanish when the symmetry is broken. The removable value is the conserved quantity.

Part II: The Complete Catalog

2.1 Number theory

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified					
1	Riemann zeta		zeta(s)	/	zeta(1?s)	zero rho		chi(rho)	= 1 iff Re(rho)=½	Probe	Yes
2	GRH Dirichlet		L(s,chi)	/	L(1?s,chi?)	zero rho		epsilon(chi)	= 1	Probe	Yes
3	BSD	L(s,E)/(s?1)^r	s=1	Leading coefficient	Probe	Yes					
4	abc conjecture	log(c)/log(rad(abc))	(1,0,1)	1	Vanishing Rate	Yes					
5	Fermat little	(a^{p?1}?1)/(a?1)	a=1	p?1	Vanishing Rate	Yes					
6	Euler product	Prod(1?p^{?s})/zeta(s)	s=1	1	Probe	Yes					
7	Weil explicit	?zeta'/zeta vs Sum log(p)/(p^s?1)	s->1	Prime identity	Probe	Yes					
8	Zeta FE	zeta(0) via FE	s=0	?1/2	Vanishing Rate	Yes					
9	PNT	pi(x)·log(x)/x	x->inf	1	Vanishing Rate	Yes					
10	Khintchine	q·	x?p/q	/psi(q)	convergents	1/sqrt5 (golden)	Vanishing Rate	Yes			
11	Möbius	(s?1)/zeta(s)	s=1	1	Probe	Yes					

2.2 Complex analysis

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
12	Argument principle	f'(z)/f(z)	zero rho	multiplicity k	Index	Yes
13	Cauchy integral	f(z)/(z?a)	zero a	f'(a)	Vanishing Rate	Yes
14	Picard little	f(z)/z^k	zero of order k	f^(k)(0)/k!	Vanishing Rate	Yes
15	Taylor remainder	R_n(x)/(x?a)^{n+1}	x=a	f^(n+1)(a)/(n+1)!	Vanishing Rate	Yes
16	FTA	f(z)/(z?z_0)^k	root z_0	g(z_0)	Vanishing Rate	Yes
17	Stirling	(n!/S?1)·n	n->inf	1/12	Vanishing Rate	Yes
18	Wallis product	Prod(2n)^2/((2n?1)(2n+1))	n->inf	pi/2	Vanishing Rate	Yes
19	Cesàro	(1?1+1?1+...) mean	N->inf	1/2	Vanishing Rate	Yes

2.3 Algebraic topology and geometry

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
20	Poincaré-Hopf	(1/2pi)?VxdV/ V ²	zero of V	index (integer)	Index	Yes
21	Atiyah-Singer	dim ker(D)?dim coker(D)	D operator	topological index	Index	Yes
22	Gauss-Bonnet	intK dA / (2pichi)	surface	1	Index	Yes
23	Riemann-Roch	l(D)?l(K?D)	divisor D	deg(D)?g+1	Index	Yes
24	Weyl law	N(lambda)/lambda^{d/2}	lambda->0+	Weyl constant	Index	Yes
25	Selberg trace	Tr(e^{?tDelta})	t->inf	1 (zero mode)	Index	Yes
26	Lefschetz	Sigma ind(p) at fixed points	fixed points	Euler characteristic	Index	Yes
27	Morse theory	f(x)/Q(x) (Hessian ratio)	critical point	±1 (Morse index)	Index	Yes
28	Sard theorem	critical values measure	critical set	0	Index	Yes
29	Stokes/de Rham	int_M domega / int_{∂M} omega	degenerate boundary	1	Index	Yes
30	Green's function	G(x,x)	diagonal	eigenfunction reciprocal	Index	Yes
31	Euler-Maclaurin	x/(e^x?1)	x=0	1 (B_0)	Vanishing Rate	Yes

2.4 Analysis and approximation

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
32	Central limit	(phi(t)?1)/t^2	t=0	?sigma²/2	Vanishing Rate	Yes
33	Rayleigh quotient	(Ax·x)/(x·x)	x=0	eigenvalue	Vanishing Rate	Yes
34	Banach fixed point	(T(x)?x)/(x?x*)	x=x*	T'(x*)?1	Vanishing Rate	Yes
35	Brouwer fixed point	(f(x)?x)/(x?x*)	x=x*	f'(x*)?1	Vanishing Rate	Yes
36	Fourier uncertainty	R(f_epsilon) = 4pisigma_xsigma_xi	epsilon->0	uncertainty bound	Vanishing Rate	Yes
37	Poisson summation	Sumf(n)/Sumf?(n)	trivial	1	Probe	Yes
38	Saddle point	g'(x)/(x?x*)	x=x*	g''(x*)	Vanishing Rate	Yes
39	Laplace method	I(n)·sqrtn	n->0	sqrtpi	Vanishing Rate	Yes
40	Noether-Landau	dF/ds	s=0	Landau coefficient	Conservation	Yes

2.5 Mathematical physics

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
41	Ising model	chi/ T?T_c ^{?gamma}	T->T_c	critical amplitude	Critical	Yes
42	Spectral gap	Delta(L)·L^z	h->1	C ~ pi	Critical	Yes
43	Lorenz attractor	log(delta)/t	t->0	lambda_1 ~ 0.91	Critical	Yes
44	Wigner semicircle	rho(lambda)/sqrt(4?lambda²)	lambda->±2	1/(2pi)	Critical	Yes
45	Semicircle N(E)/E	N(E)/E	E->0	1/pi	Vanishing Rate	Yes

2.6 Information theory and statistics

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
46	Shannon entropy	p·log(p)	p->0	0	Critical	Yes
47	Boltzmann entropy	S/ln(W)	W=1	1	Critical	Yes
48	Bayes theorem	P(H D) as P(D)->0	D impossible	P(H) (prior)	Conservation	Yes
49	Fourier uncertainty	sigma_x·sigma_xi for f_epsilon	epsilon->0	constant (bound)	Vanishing Rate	Yes

2.7 Optimization and control

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
50	Gradient descent	Deltatheta/eta	eta->0	??L	Conservation	Yes
51	KKT conditions	mu_i/g_i(x*)	active constraint	shadow price lambda	Conservation	Yes
52	Schanuel	e^{alpha_1}/e^{alpha_2}	alpha_1=alpha_2	1	Conservation	Yes
53	Noether theorem	dL/depsilon	epsilon=0	conserved quantity	Conservation	Yes

2.8 Algebra

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
54	Pythagorean theorem	$a^2+b^2?c^2$ at right angle	theta=pi/2	0 (with c=hypotenuse)	Conservation	Yes

2.9 Dynamical systems

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
55	Poincaré recurrence	epsilon·tau(epsilon)	epsilon->0	constant	Critical	Yes

Part III: Cross-References

3.1 By removable value type

Removable Value Type	Experiments	Count
Integer	Poincaré-Hopf, Argument principle, Atiyah-Singer, Morse, Fermat little, FTA	6
Rational (simple)	Euler-Maclaurin (1), Cesàro (1/2), Taylor (f^(n+1)/(n+1)!), Stirling (1/12), Wal	8
Real (transcendental)	Zeta FE (?1/2), CLT (?sigma^2/2), Wigner (1/(2pi)), Laplace (sqrtpi)	4
1 (identity)	Riemann zeta, GRH, Euler product, Poisson, Schanuel, Stokes, Green, Gauss-Bonnet	8
0	Shannon, Boltzmann (S/ln(W)), Sard, Morse saddle	4
Function of parameters	BSD (rank+Sha), Ising (C), Spectral gap (C), Lorenz (lambda), Khintchine (1/sqrt	13
Nonexistent (not removable)	-- (all 55 are removable)	0

3.2 By mathematical domain

Domain	Experiments	Mechanisms Used
Number theory	1-11	Probe (6), Vanishing Rate (5)
Complex analysis	12-19	Index (2), Vanishing Rate (6)
Algebraic topology/geometry	20-31	Index (11), Vanishing Rate (1)
Analysis/approximation	32-40	Vanishing Rate (7), Probe (1), Conservation (1)
Mathematical physics	41-45	Critical (3), Vanishing Rate (2)
Information theory	46-49	Critical (2), Conservation (1), Vanishing Rate (1)
Optimization	50-53	Conservation (4)
Algebra	54	Conservation (1)
Dynamical systems	55	Critical (1)

3.3 By mechanism

Mechanism	Experiments	Count	Character
Probe	1, 2, 3, 6, 7, 37	6	Tests identity of two objects
Index	12, 20-31	13	Extracts integer (winding, multiplicity)
Vanishing Rate	4, 5, 8, 9, 10, 11, 15-19, 32-36, 38-40, 45, 49	2[illegible]	Leading Taylor coefficient
Critical	41-43, 46, 47, 55	6	Phase transition / critical amplitude
Conservation	48, 50-54	6	Conserved quantity from symmetry

Part IV: The Decision Tree

Given a suspected 0/0 form f/g at a point x?:

```
Step 1: Verify f(x?) = g(x?) = 0
  If not -> not a 0/0 form
```

```
  If yes -> proceed

Step 2: Compute the limit lim_{x->x?} f(x)/g(x)
  If limit does not exist -> not a removable singularity
  If limit exists and is finite -> removable value found
  If limit is inf -> pole (not 0/0 in the useful sense)

Step 3: Classify the removable value
  Is it an integer?
    -> Yes: INDEX mechanism (topological invariant, multiplicity)
    -> No: proceed

  Is f/g = 1 where defined (before taking the limit)?
    -> Yes: PROBE mechanism (tests functional equation)
    -> No: proceed

  Does the removable value depend on a rate of vanishing?
    -> Yes: VANISHING RATE mechanism (Taylor coefficient)
    -> No: proceed

  Is the 0/0 at a phase transition?
    -> Yes: CRITICAL PHENOMENON mechanism
    -> No: proceed

  Does the 0/0 arise from a symmetry?
    -> Yes: CONSERVATION mechanism
    -> No: Unknown mechanism -- classify it

Step 4: Verify computationally
  Compute f/g numerically near x?
  Check convergence to the removable value
  Compare with the theorem's prediction
```

Part V: The Connections

5.1 How the mechanisms relate

The five mechanisms are not independent. They form a hierarchy:

```
CONSERVATION <-?? creates ??-> PROBE
    ?                          ?
    ?                          ?
    ?                          ?
VANISHING RATE <-?? specializes ??-> INDEX
    ?
    ?
    ?
CRITICAL PHENOMENON
```

Conservation creates Probe: A conserved quantity (Noether) implies a functional equation (zeta FE), which implies f/g = 1 where defined (the Probe). The Probe is the remnant of the conservation law.

Vanishing Rate specializes to Index: When the removable value is an integer, the Vanishing Rate mechanism specializes to the Index mechanism. The integer is the winding number, the multiplicity, or the topological invariant.

Critical Phenomenon is the physics of Vanishing Rate: At a phase transition, the order parameter and susceptibility both vanish (or diverge) at the critical point. Their ratio is a 0/0 whose removable value is the critical amplitude -- the universal constant that characterizes the universality class.

5.2 The unified picture

All five mechanisms share a common structure:

1. Two objects vanish at a point
2. Their ratio is 0/0
3. The removable value encodes structure

The difference is what kind of structure:

- Probe: structural identity (is f = g?)
- Index: topological invariant (what is the winding number?)
- Vanishing Rate: analytic invariant (what is the leading coefficient?)
- Critical: physical invariant (what is the critical amplitude?)
- Conservation: symmetry invariant (what is the conserved quantity?)

All five are invariants -- quantities that do not change under perturbation. The 0/0 form is the mechanism by which these invariants are extracted from the mathematics.

Part VI: Open Problems

6.1 Missing experiments

Branch	Potential 0/0	Status
Galois theory	Discriminant of polynomial at repeated root	Not computed
Algebraic K-theory	Bott periodicity at degenerate spectra	Not computed
Geometric analysis	Ricci flow at neck pinch	Not computed
Category theory	Natural transformation at degenerate object	Not computed
Combinatorics	Generating function singularity at radius of convergence	Partially explored
Ergodic theory	Birkhoff average at exceptional points	Not computed
Functional analysis	Spectral density at band edge (general)	Partially explored
Arithmetic geometry	Néron-Tate height at torsion points	Not computed

6.2 Classification completeness

Is the five-mechanism classification complete? We conjecture it is, based on the following argument:

Every 0/0 form f/g tests the relationship between f and g at x?. The possible relationships are:

1. f = g (Probe)
2. f/g winds around x? an integer number of times (Index)
3. f and g have specific leading coefficients (Vanishing Rate)
4. f and g diverge at a phase transition (Critical)
5. f and g are constrained by symmetry (Conservation)

We believe these five cover all possible relationships between two vanishing functions. A proof would require showing that every analytic 0/0 form falls into one of these categories.

6.3 The constructive problem

Can the 0/0 principle be used to discover new theorems? The approach would be:

1. Construct a 0/0 form f/g in a new setting
2. Compute the removable value numerically
3. If the removable value is "interesting" (an integer, a known constant, a simple expression), search for a theorem that explains it
4. If the removable value is "new" (not previously known), it may itself be a theorem

This is the constructive version of the 0/0 principle: not just verifying known theorems, but using 0/0 forms to find new ones.

Appendix: Data Files

Each experiment produces a JSON data file in data/. The files contain:

- The 0/0 form and its parameters
- Numerical values of the removable value at multiple points
- Convergence data (how the removable value is approached)
- The summary verdict (SUPPORTED / NOT SUPPORTED)
- Honest wall statements (limitations of the numerical verification)

Experiment	**Data File**
Riemann zeta	data/argument_principle_0_over_0_data.json
GRH Dirichlet	data/grh_dirichlet_0_over_0_data.json
BSD	data/bsd_0_over_0_data.json
abc	data/abc_conjecture_0_over_0_data.json
Fermat little	data/fermat_little_0_over_0_data.json
Euler product	data/euler_product_0_over_0_data.json
Weil explicit	data/weil_explicit_0_over_0_data.json
Zeta FE	data/zeta_functional_eq_0_over_0_data.json
PNT	data/prime_number_theorem_0_over_0_data.json
Khintchine	data/khintchine_0_over_0_data.json
Möbius	data/mobius_function_0_over_0_data.json
Argument principle	data/argument_principle_0_over_0_data.json
Cauchy integral	data/cauchy_integral_0_over_0_data.json
Picard	data/picard_little_0_over_0_data.json
Taylor	data/taylor_remainder_0_over_0_data.json
FTA	data/fta_0_over_0_data.json
Stirling	data/stirling_approx_0_over_0_data.json
Wallis	data/wallis_product_0_over_0_data.json
Cesàro	data/cesaro_summation_0_over_0_data.json
Poincaré-Hopf	data/poincare_hopf_0_over_0_data.json
Atiyah-Singer	data/atiyah_singer_0_over_0_data.json
Gauss-Bonnet	data/gauss_bonnet_0_over_0_data.json
Riemann-Roch	data/riemann_roch_0_over_0_data.json
Weyl law	data/weyl_law_0_over_0_data.json
Selberg trace	data/selberg_trace_0_over_0_data.json
Lefschetz	data/lefschetz_fixed_point_0_over_0_data.json

Morse theory	data/morse_theory_0_over_0_data.json
Sard	data/sard_theorem_0_over_0_data.json
Stokes/de Rham	data/stokes_de_rham_0_over_0_data.json
Green's function	data/greens_function_0_over_0_data.json
Euler-Maclaurin	data/euler_maclaurin_0_over_0_data.json
CLT	data/central_limit_theorem_0_over_0_data.json
Rayleigh	data/rayleigh_quotient_0_over_0_data.json
Banach	data/banach_fixed_point_0_over_0_data.json
Brouwer	data/brouwer_fixed_point_0_over_0_data.json
Fourier uncertainty	data/fourier_uncertainty_0_over_0_data.json
Poisson summation	data/poisson_summation_0_over_0_data.json
Saddle point	data/saddle_point_0_over_0_data.json
Laplace	data/laplace_method_0_over_0_data.json
Noether-Landau	data/noether_landau_0_over_0_data.json
Ising	data/ising_model_0_over_0_data.json
Spectral gap	data/spectral_gap_0_over_0_data.json
Lorenz	data/lorenz_attractor_0_over_0_data.json
Wigner	data/wigner_semicircle_0_over_0_data.json
Shannon	data/shannon_entropy_0_over_0_data.json
Boltzmann	data/boltzmann_entropy_0_over_0_data.json
Bayes	data/bayes_theorem_0_over_0_data.json
Gradient descent	data/gradient_descent_0_over_0_data.json
KKT	data/kkt_conditions_0_over_0_data.json
Schanuel	data/schanuel_0_over_0_data.json
Noether	data/noether_theorem_0_over_0_data.json
Pythagorean	data/pythagorean_0_over_0_data.json

This atlas is a reference document. For the philosophical interpretation, see ON_THE_NATURE_OF_ZERO.md. For the synthesis, see THE_UNIVERSAL_ZERO.md. For the epistemology, see REMOVABLE_SINGULARITIES.md.

All 55 experiments verified computationally. 149 regression tests passing.